Detection Fusion of YOLO and RT-DETR for Unique Coral Instance Estimation

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Abstract*- Coral reef monitoring is very important for marine ecosystem but the current methods take a lot of time, effort and are costly. To make this easier, we created a deep learning-based detection system that can watch underwater videos and find coral automatically. The system uses YOLOv8 and RT-DETR object detection models to automatically detect and monitor coral reef objects from underwater images and videos. We trained the models using the AIMECORAL1 dataset containing 580 images collected from the Southwest Indian Ocean and the AIMECORAL2 dataset containing 282 images collected from New Caledonia in the Pacific Ocean. The performance of the models was evaluated using standard object detection metrics such as Precision, Recall. The YOLOv8 model achieved a Precision of 87.15%, Recall of 83.28%. The RT-DETR model achieved a Precision of 86.21%, Recall of 87.96%, showing better overall detection and localization performance compared to YOLOv8. Our proposed fusion detection and tracking model was implemented to improve temporal tracking consistency in underwater videos. It achieved a Precision of 53.09% and a Recall of 97.66%. The experimental results show that the fusion approach improves object tracking stability across frames. Tracking consistency was evaluated as the number of stable IDs and average track length which served as measures of how consistently individual coral instances were tracked across frames. The proposed system reduces manual effort, lowers monitoring cost, and provides an efficient solution for automated coral reef monitoring and conservation. This work demonstrates how artificial intelligence can support sustainable marine ecosystem conservation through automated coral reef monitoring.

Keywords: Coral Reef Monitoring, YOLO, Object Detection, Multi-Object Tracking, DeepSORT, Sustainability

I. Introduction

Coral reefs are vast underwater features created by colonies of small animals called coral polyps. These are commonly described as “rainforests of the sea”. They cover 0.1% of the ocean bottom yet provide a habitat for 25% of marine species. Coral reefs are very important for different kinds of sea life as they provide critical ecosystem services like food supply, coastal protection and tourism income. They are very fragile and can be easily damaged by change in climate, pollution and ocean acidification. Because these problems are getting worse, we need to check coral reefs more carefully are observe regularly; this helps us keep them safe. To prevent further damage, we must take action quickly, use proper information and save coral reefs.

Traditional reef observation systems depend on manually collected information by people who go into the water and monitor the reefs. They collect data such as the coral health, area covered and types. They do this by scuba diving and take pictures to study later. Problems with these methods are they are expensive and need a lot of time and can cover only small areas and the results may not be fully accurate as they depend on the person who is observing. Underwater images are difficult because of poor lighting and difficult to find partially hidden or overlapping corals. So, some may miss corals or lose them during tracking.

To address these challenges, this study outlines a deep-learning based system that monitors corals continuously, tracks them, and estimates unique corals in underwater videos or images. Our model uses advanced object-detection systems YOLOv8 and RT-DETR. These models automatically detect corals in underwater videos and images. YOLOv8 (You Only Look Once) is CNN-based object-detection model designed by Ultralytics, developed for high accuracy and speed. It can run on simple computers, and provides more accuracy on powerful systems. This type is faster and more accurate than previous YOLO versions. It has better design that helps understand the images clearly and is easy to use. It can also run on many devices because it comes in with model sizes.

RT-DETR (Real-Time Detection Transformer) is a high-level model made by Baidu that can detect objects in images very fast and accurately. It finds objects quickly and in a smarter +way than older models like YOLO. It is more stable as it does not need extra steps to remove duplicate boxes, has better input processing as it looks at images in different sizes which helps in understanding objects clearly. It uses most useful information which improves accuracy and allows us to adjust speed without needing us to train again.

The models individually have strengths but can still miss small or difficult corals. So, the fusion of yolo and RT-DETR using WBF (Weighted Box Fusion) combines detections from both models, if both models detect same coral they are merged, and if one model detects an object while the other misses it, it is still included. The WBF is an object detection ensemble method which combines boxes from multiple models and creates one better final box instead of removing repeated boxes. Overall, the fusion helps recover additional corals that may be missed.

ByteTrack is a system that keeps track of objects in videos and images. It almost tracks all detected objects, not just the ones that are easy to notice. It follows the same object correctly even when there are many objects close together. This assigns id's to detected corals so, we can estimate how many unique corals exist and how long each coral stays tracked and whether the IDs remain stable. Maintaining the consistent object identities across consecutive frames, ByteTrack can reduce the identity switches and broken paths in long video sequences. By blending environmental domain knowledge with AI helps us check coral reefs better and protect them.

II. LITERATURE SURVEY

In this work the author used a vision foundation model and fine-tuned it with adapters for analysis of images even if they belong to different times and locations but it requires a lot of computation power [1]. This paper suggests using YOLO based models for creating an automatic soft coral detection system but it has limited generalization and achieves precision of 84.6% [2]. This paper uses Autonomous Underwater Vehicles (AUVs) which are combined with multi sensor system and AI vision for monitoring the corals. This fails in navigation, lacks standard methods and is costly [3]. The author uses satellite data with CNN and logistic regression models for coral reef detection, but the performance is highly dependent on data quality and doesn't perform well in complex ocean regions [4]. The system tries to understand coral reefs from videos. It uses YOLOv5 and stereo vision for detection of coral. It's difficult to use this in unclear water but still it manages to achieve good results [5]. The author of this paper has proposed the idea of using smaller deep learning network to detect corals as it is faster and can be efficient. Being lightweight, it does miss complex patterns [6]. The focus of this research is on finding unhealthy corals. It uses EfficientNet for the same. It doesn't perform well while detecting small or hidden corals [7]. This paper uses underwater cameras as well as drones for studying corals. It uses CNN for feature extraction. It combines underwater images and aerial images for classifying the reefs. It requires multiple devices and complex data integration but has achieved great results [8]. The idea behind this paper is to use ResNet50(CNN) model for classifying images of corals based on their health. Dataset quality and balancing play an important role in this paper and is also its biggest limitation. Despite this the paper has achieved good result [9]. This paper is focused on recognizing different coral species using computers. Uses CNN trained on underwater images. It is difficult for CNN as corals look similar and have complex structure [10]. This work is focused on studying coral textures for species identification. It uses ResNet with transfer learning and data augmentation. The problem with this paper is that a small dataset and class imbalance affect its performance [11]. This research work classifies coral health into different categories. It uses two stream CNN model for detecting different types of corals. It requires large labelled datasets and struggles due to high computation requirement [12]. This paper focuses on detection of underwater objects like corals. It uses YOLOv8 based Underwater Detection Network (UDON). It is challenged by complex underwater scenes and poor visibility and achieves an mAP50 of 84% [13]. The aim of the authors of this paper is to detect corals even when labelled data is limited. They use semi supervised deep learning segmentation to identify coral regions in images. They have implemented it in Pujada Bay, Philippines but this research is highly dependent on unlabelled data quality and may not work well every time [14]. This paper uses a system for automatically watching and identifying corals using videos. It uses YOLOv8 for coral detection. It needs a lot of labelled data and finds it difficult when the water is blurry. It achieves precision of 84.7% [15]. The literature survey shows that existing methods mainly use single deep learning models for coral detection and monitoring. Therefore, this work proposes a detection fusion framework with YOLO, RT-DETR and ByteTrack for better coral detection and unique coral instance estimation.

III. PROPOSED METHODOLOGY

A. *Overall Pipeline*

We have underwater images and we want to find where are corals and how many distinct corals are there. We have two datasets AIMECORAL1 (580 images from the Southwest Indian Ocean) and AIMECORAL2 (282 images from New Caledonia, Pacific Ocean). We teach the models what corals look like by fine tuning.

Two detection models are used - YOLOv8 and RT-DETR. YOLO is very cautious and has fewer false detections, but may miss few corals. While RT-DETR finds more corals and better in overall detection. Now we combined both models and created a fusion model for complete detection. The fusion model works as follows: when both models detect a coral, we keep it. Corals missed by RT-DETR and found by YOLO are also added. Hence, very less corals are missed.

To keep track of unique corals, we used a tracking system ByteTrack, which assigns unique and stable Ids to each coral, and keeps the Ids across frames, which helps us reduce counting same coral multiple times. We can estimate total unique corals by counting the stable Ids.

The complete framework, as shown in Fig.1, is composed of five major stages. In the first stage, the input is made using the underwater coral images from the AIMECORAL1 and AIMECORAL2 datasets. Second, both YOLOv8 and RT-DETR models are trained by a two-stage domain adaption approach, where the models are first trained on AIMECORAL1, and then fine-tuned on AIMECORAL2. Both models are trained for the individual detection of coral areas and return bounding boxes. Then the detection results are fused by Weighted Box Fusion (WBF) to produce a single refined detection output. Finally, the fused detections are passed to ByteTrack tracking algorithm that assigns unique IDs to each detected coral and keeps track of these IDs in consecutive frames. The total number of unique coral instances is then estimated with stable tracking IDs. The details of each stage are described in the following subsections.

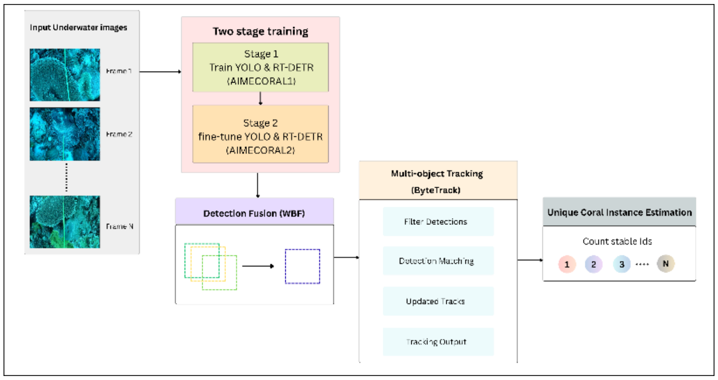

Fig.1. Overall detection-fusion-tracking pipeline

B. *Domain adaption*

Corals do not look the same everywhere. When a model is trained to detect corals from one place it learns how corals look in that environment like colour and shape, but in another environment, they might look different because of lighting or water conditions. They are still corals, but this might confuse the model and might decrease detection accuracy. This system is first trained on AIMECORAL1 dataset where it learns how corals and their structures look like. This trained model is again trained on AIMECORAL2 dataset where it adapts to different conditions and can detect different types of corals. Here the model learns from one domain and adapts to another as illustrated in Fig.2.

The two-stage domain adaptation is carried out in a structured manner. In Stage 1, both the YOLOv8 and RT-DETR models with pretrained weights are trained using the AIMECORAL1 dataset. In this stage, both the models will learn about the coral's overall shape, texture, and contours. Once the stage 1 ends, the best performing model is selected, and its weight are stored. In Stage 2, the stored weights are used and fine-tuned using AIMECORAL2 dataset instead of training the algorithm from scratch again. This fine-tuning allows both models to adapt the features learned from first dataset to the different underwater conditions present in the second dataset. As a result, the models are able to retain what they have learned about source domain while simultaneously adapting to the target domain, which improves the detection performance.

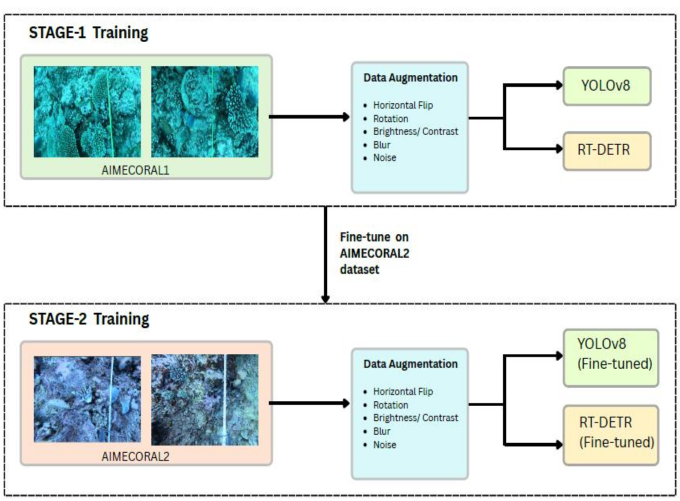

Fig.2. Two-stage domain adaptation through Stage 1 training on AIMECORAL1 followed by fine-tuning on AIMECORAL2

C. *Fusion Model*

Fusion model consists of two detectors - YOLO and RT-DETR, where detection results from both detectors are fused together to improve detection completeness. Single detectors have some limitations. They often find it hard to detect small objects clearly. They also find it difficult when objects are blocked by other things. Detection becomes even more difficult underwater as the images have low contrast and lighting is poor. Because of these challenges, some objects may be missed by single detectors or give less accurate results.

Different models behave in different ways, and depends on the approach used produce different predictions accordingly. For example, YOLO is more careful and very selective when detecting objects. Usually, this model chooses correctness and makes fewer false detections to avoid incorrect detections (false positives). As a result, such model will comparatively have high precision because almost all detected objects are correct, but may miss some objects.

On the other hand, RT-DETR is quite opposite. It behaves in more aggressive way. It tries to maximize the number of detections, even if it is not fully sure about all of them. So, RT-DETR has high recall, as it finds most objects, but sometimes gives incorrect results. Overall, YOLO is more careful and tries to avoid mistakes, while RT-DETR aims to detect as much as possible, even if it means making a few extra errors. The fusion approach provides a stable solution by combining advantages of various models and improves the detection completeness and reduce missing corals. However, it may lead to more false detections.

The fusion approach used in this study is inference-level fusion. This means two models are trained separately, where each model learns on its own. After both models are fully trained, they are combined at the prediction stage and not during training stage. When a new image or a video frame is given, both models first make their own predictions after that their outputs (bounding boxes and detections) are merged together using a fusion method called WBF (Weighted Box Fusion). So, fusion happens only during final step and not during learning as shown in Fig.3. This approach is important because it makes the system very easy to use. Since the models are trained separately, so there is no need to re-train them together again, which saves a lot of time and effort. If one model is updated, we can replace it without affecting the other model. Another advantage is flexibility, where you can fuse different models such as YOLO and RT-DETR even though they have different architectures. The models also remain independent since; each comes with its advantages that should not be ignored. For example, if one fails to detect a coral, another model may detect it. Fusing models on inference level makes process easier, flexible, efficient and effective.

During the fusion, the main focus is on combining bounding boxes and corresponding confidence scores from both models. The fusion model compares boxes from YOLO and RT-DETR that are pointing to the same coral even if shifted slightly or in different size. Later it merges into one bounding box and adjusts the confidence scores. Overall, the system is carefully merging the location information (bounding box) and certainty level (confidence score), from both detectors for better final result. Since this is coral detection, the class labels are always same and need not be compared. This makes fusion simple, because it only focusses on improving where the coral is and how confident the system is.

Weighted Box Fusion (WBF) combines the predictions of several models into one prediction through the fusion of the box predictions, unlike the non-maximum suppression (NMS) that just removes duplicate boxes. NMS works by filtering out the overlapping boxes and only keeps those with highest probability prediction. Though, this approach removes duplicates, it may also remove some useful information by removing multiple boxes that are actually used to identify the same coral. Here, WBF does not delete the boxes, instead it merges them into one which helps in preserving important information from both the models and calculates an average weight based on the coordinates and their confidence scores. This improves the overall performance of object detection.

First, each bounding box is normalized between 0 and 1. The WBF uses IOU (Intersection Over Union) score to check which two boxes are overlapping. The IoU is used to determine if the overlapping boxes refer to the same coral or not. IoU checks how much two boxes match with each other. If the overlap between two boxes is high then the IoU will also be high, which means that these boxes refer to one type of coral.

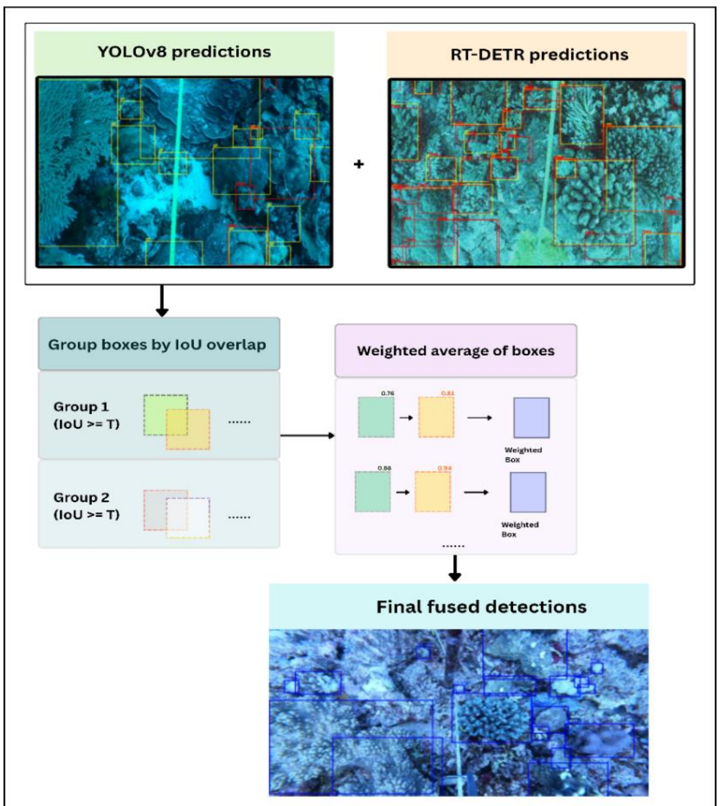

Fig. 3. Weighted Box Fusion (WBF) combines the predictions from YOLOv8 and RT-DETR.

When processing fusion, not all models contribute equally. In our case, RT-DETR contribution is higher than that of YOLO, as RT-DETR generates more accurate predictions. Hence, the output results depend more on RT-DETR's results. This shows that fusion is not random, based on how each model operates. Confidence thresholding is also applied before performing the fusion, this means that the boxes with very low confidence are removed as they may increase noise. Filtering them out helps in improving overall results.

The confidence values for the two detectors used in the fusion process were determined based on the performance of each model individually. When comparing RT-DETR with YOLOv8, RT-DETR outperformed YOLOv8 in terms of recall. As a result, RT-DETR was given more weight during the fusion process, and YOLOv8 was used as a complementary detector to identify corals that were not detected by RT-DETR. The final detection output becomes more complete without relying entirely on a single detector.

By combining detections of two models, we are able to improve recall, means improved chance of finding all corals fewer missed corals. On the other hand, precision might decrease slightly because adding detections may lead to extra or false positives. This trade-off between precision and recall, in our case, improving detection completeness is more important for accurate coral monitoring. Moreover, fusion improves tracking efficiency by preventing track break caused by missed detections, meaning the same coral might be counted multiple times or lost temporarily. The fusion reduces this, and becomes consistent in tracking objects. This improves tracking reliability and reduces counting errors.

The purpose of fusion is to improve the overall performance of the system. Better detection leads to better tracking, and better tracking leads to more accurate coral count estimation. This improves the overall performance of the coral monitoring model.

The proposed fusion method involves computations greater than those involved in using only one object detection model since YOLOv8 and RT-DETR independently perform inference on the same image. Weighted Box Fusion combines the overlapping bounding boxes to create a single detection result after the two models generate detections. ByteTrack then uses the detections to perform object association between consecutive frames. As a result, the proposed framework requires additional computation due to the execution of two detection models followed by the fusion and tracking processes.

D. *Tracking and Coral Estimation*

Tracking means following same coral across many frames. If the camera moves, the same corals may look slightly shifted, and without tracking the system may mistake them for new corals instead of recognizing them as the same ones. So, this assigns a unique Id to each coral. ByteTrack tries to keep same Id for same coral. It looks at position and continues detection to ensure consistency.

Here, total Id are number of unique corals. Stable Ids are Ids lasting many frames. If an Id appears only once and disappears then it is noise. It's a real coral if the Id stays for many frames. How long the Is states consistent is the average track length.

In this study, an ID is considered stable when it lasts for at least five consecutive frames. This cut-off point has been selected as a reasonable standard to distinguish between false ones or temporary observations and persistent ones. IDs lasting for one or two frames are more prone to being false detections or temporary error in the tracking process; hence, IDs that last more than five frames are more likely to correspond to real corals.

Tracking consistency is calculated using the number of stable IDs and the average track length. A higher number of stable IDs show that the tracker can maintain the same identity for a larger number of corals without generating new identities often. A higher average track length indicates that the same coral can be tracked continuously across multiple consecutive frames. The combination of these two factors supports the evaluation of tracking consistency.

During tracking, identity switches may occur due to missed detections; the ID of the same coral can be switched in subsequent frames. Similarly, when a coral is missed for a few consecutive frames, the path becomes fragmented, and the number of tracks increases instead of having a single track. The suggested fusion model can solve these problems as there would be more complete detections because of which ByteTrack receives continuous detections and thus can maintain the same ID across consecutive frames. The tracking performance comparison using Total IDs, Stable IDs and Average Track Length is shown in Fig. 4.

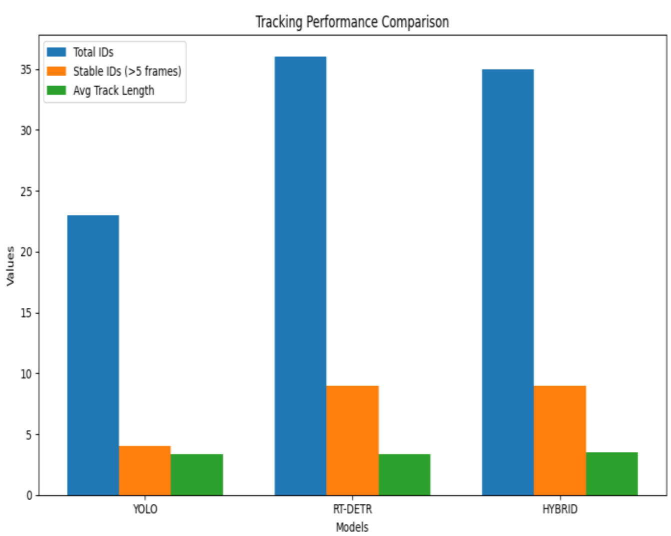

Fig. 4. Comparison of tracking performance between YOLO, RT-DETR, and the proposed Fusion model.

IV. Results and Discussions

The performance of the coral reef monitoring system proposed in this paper was evaluated using YOLO, RTDETR, and Fusion model. We evaluated using established metrics for evaluating system's performance: Precision, Recall.

A. *YOLO*

The detection model which is based on YOLO achieved a Precision of 87.15% and a Recall of 83.28%.

From the results achieved it can be said that YOLO can detect coral reef objects with high precision. The model can perform real time detection and identify coral structures.

A qualitative example of the coral detections generated by the YOLOv8 model and the corresponding ground-truth annotations is illustrated in Fig. 5.

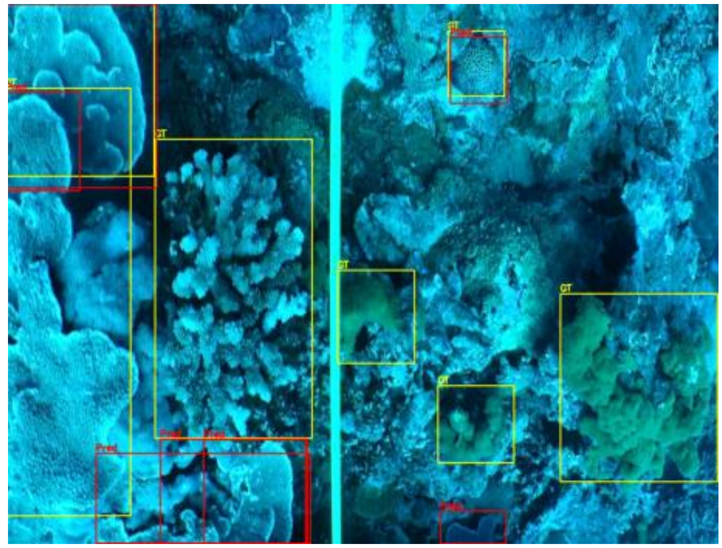

Fig. 5. Qualitative detection results of the YOLOv8 model. The red bounding boxes indicate the detected corals, while the yellow bounding boxes represent the ground-truth annotations.

B. *RT-DETR*

The detection model which is based on RT-DETR achieved a Precision of 86.21% and a Recall of 87.96%.

The RT-DETR model performs better then YOLO based model. The recall value has increased which suggests that a larger number of coral reefs can be detected with fewer missed detections.

Fig. 6 shows a qualitative example of the coral detections produced by the RT-DETR model along with the corresponding ground-truth annotations.

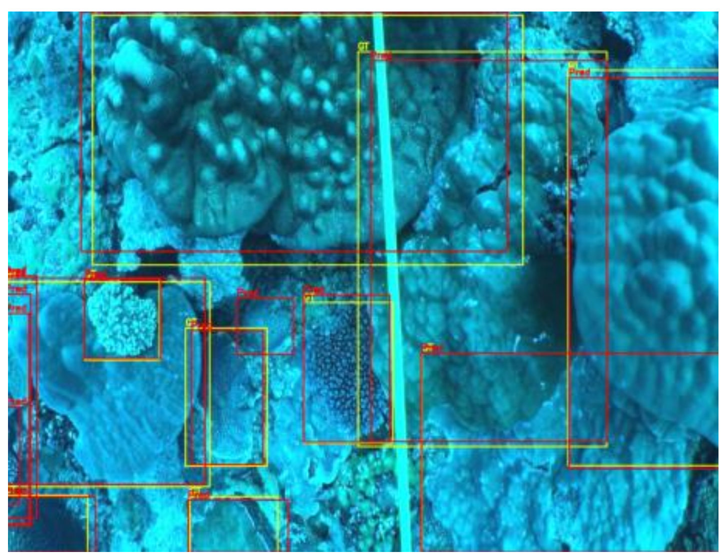

Fig. 6. Qualitative detection results of the RT-DETR model. The red bounding boxes indicate the detected corals, while the yellow bounding boxes represent the ground-truth annotations.

C. *Proposed Fusion Model*

The Proposed Fusion model achieved a Precision of 53.09% and a Recall of 97.66%.

The exceptional high recall of fusion model (as shown in Fig. 7 and Fig. 8) says that it can detect almost all coral reef objects with very few missed detections. This is very important as missed corals can affect the assessment accuracy.

In addition to that, the proposed fusion framework has been tested qualitatively under different underwater conditions such as changes in lighting, water clarity and coral density. According to the visual observations, it was concluded that the proposed fusion framework is capable of detecting coral instances in difficult situations when the underwater environment was not clearly visible or well-lit. Though there were few false detections, it was evident that the proposed model detected more coral instances than individual detectors.

Table 1

PRECISION AND RECALL COMPARISON

Model	**Precision**	**Recall**
YOLO	0.8715	0.8328
RT-DETR	0.8621	0.8796
Fusion Model	0.5309	0.9766

The comparison results show that the proposed Fusion model achieves the highest recall. It can be used for the studies which need high sensitivity. The proposed system shows it can automate coral reef monitoring and replace or assist manual ecological surveys.

D. *Tracking Performance Analysis*

The performance of tracking was evaluated using metrics such as Total IDs generated, Stable IDs, and Average Track Length.

Total IDs represent the total number of unique object identities generated during tracking.

Stable IDs (>5 frames) indicate the number of tracked objects that remained consistently detected for more than five consecutive frames.

Average Track Length represents the average duration for which an object was successfully tracked throughout the video sequence.

Table 2

TRACKING PERFORMANCE

Model	**Total IDs**	**Stable IDs (>5 Frames)**	**Average Track Length**
YOLO	23	4	3.35
RT-DETR	36	9	3.39
Fusion Model	35	9	3.51

The proposed Fusion Model achieved the highest average track length of 3.51 frames while maintaining 9 stable IDs. This suggests that the integration of ByteTrack improved the persistence and continuity of tracked coral reef objects across frames. The proposed Fusion framework effectively reduced loss of identity and improved tracking stability in challenging underwater conditions.

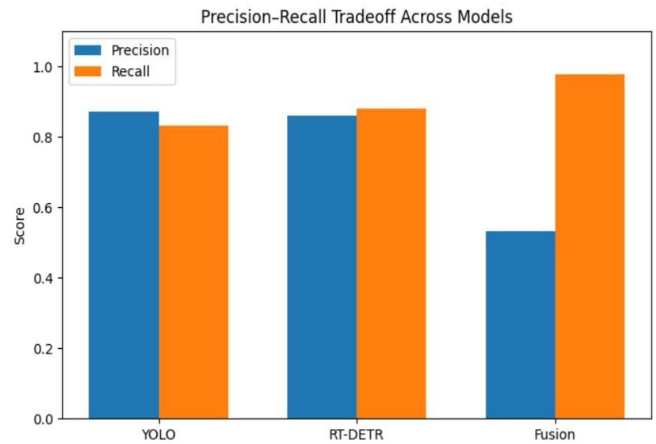

Fig. 7. Precision and recall comparison of YOLO, RT-DETR, and the proposed Fusion model.

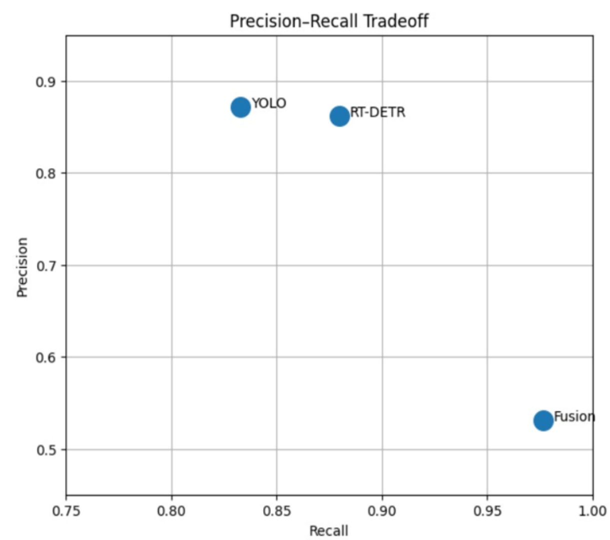

Fig. 8. Precision-recall tradeoff of YOLO, RT-DETR, and the proposed Fusion model.

V. CONCLUSION AND FUTURE WORKS

In this work, a system for coral detection and tracking from underwater images was developed. Detecting and tracking of corals is difficult because of poor lighting, water clarity, same backgrounds and irregular shape of corals. So, two object detectors, YOLO and RT-DETR are used for training on two stages. Initially the models are taught to recognize the models and at second stage they were fine-tuned.

Performance of both the models are observed where YOLO has high precision and low recall, on the other hand RT-DETR performs better with high recall. Based on this a Weighted Box Fusion has been introduced to combine detections from both models. The fusion model is introduced to reduce the number of missed corals during detection by combining advantages of both models. Thus, even when one model loses precession because of additional detections, recall is increased.

ByteTrack was applied to track the detected coals across frames and give them unique IDs to estimate unique coral instances then counting the same coral many times. Tracking results show that fusion model maintains stability and maintains longer track duration, which is useful for coral monitoring applications.

In conclusion, this work provided a practical approach for detection and tracking of coral. Though fusion model has not been able to beat RT-DETR in precision alone, it improves detection completeness and continuity in tracking. This work shows how artificial intelligence can contribute to sustainability by helping protect coral reefs through efficient and automated monitoring. This research may be improved in future through use of more advanced approaches to fusion, tracking and tracking ground truths.

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